Adaptive Behavioral History Management for Resource-Efficient Short-Video Recommendation

1 Background and Motivation

Short-video recommendation systems rely heavily on users' historical interactions to predict their preferences and select relevant content. In general, providing more behavioral history to a recommendation model may provide richer information about user preferences. However, processing longer histories also introduces additional computational overhead, including data access, preprocessing, feature construction, memory usage, and model inference cost.

This creates a trade-off between recommendation quality and computational cost. Using the complete behavioral history may not always be necessary, especially when additional historical interactions provide only a small improvement in predictive performance.

From a data-management perspective, this raises an important question: rather than always processing the maximum amount of available behavioral data, can the system determine how much history is actually useful for a given recommendation task?

This project is motivated by the EDAM (Energy-aware DAta Management) direction, which focuses on making data-management decisions with consideration for computational and resource efficiency. The project also takes inspiration from the idea of early filtering and adaptive control presented in the BLOOM-FaaS work. In BLOOM-FaaS, unnecessary data is filtered before expensive processing, while runtime information is used to adapt processing decisions. Importantly, the paper also demonstrates that an adaptive filtering mechanism is not always beneficial: when the filtering overhead exceeds its benefit, applying it may increase the overall cost.

Following this principle, this project investigates whether behavioral history can be managed adaptively so that additional historical data is processed only when its expected benefit justifies its computational cost.

2 Research Question

Main Research Question
How can behavioral history be managed adaptively to optimize the trade-off between recommendation quality and computational cost?

Vietnamese: Làm thế nào để quản lý lịch sử hành vi một cách thích ứng nhằm tối ưu sự đánh đổi giữa chất lượng gợi ý và chi phí tính toán?

The main research question can be decomposed into four sub-questions:

RQ1. Effect of behavioral history: How does the amount of behavioral history affect recommendation quality?
This establishes the relationship between history length and recommendation performance.

RQ2. Computational cost: How does the amount of behavioral history affect computational cost?
This investigates how increasing the amount of historical data affects processing requirements, such as inference time, memory usage, and the amount of data processed.

RQ3. Cost-benefit condition: Under what conditions does additional behavioral history provide sufficient benefit to justify its computational cost?
This question is motivated particularly by the BLOOM-FaaS principle that data reduction or filtering should be applied when its benefit outweighs its overhead.

RQ4. Adaptive management: Can an adaptive behavioral-history management strategy reduce computational cost while maintaining an acceptable level of recommendation quality?
This is the final system-level question: whether the observations from RQ1–RQ3 can be converted into an adaptive data-management strategy.

3 Objectives

The project has four main objectives:

Objective 1 – Quantify the value of behavioral history

Measure how recommendation performance changes as different amounts of recent behavioral history are provided to the model. The experiment will consider several history lengths:

$$K \in \{0, 1, 5, 10, 20, 50, 100, \text{Full}\}$$

where K represents the number of recent behavioral interactions used by the recommendation model.

Objective 2 – Quantify the computational implications

Measure how increasing K affects computational requirements, including:

- amount of behavioral data processed;
- inference time / latency;
- memory usage;
- model workload;
- and, where reliable measurement infrastructure is available, energy consumption.

Energy measurement will be treated as an additional evaluation dimension rather than a mandatory requirement if the experimental environment does not support reliable measurement.

Objective 3 – Identify the cost-benefit relationship

Analyze the marginal benefit obtained from additional behavioral history against the additional computational cost.

Instead of simply asking whether a longer history produces higher AUC, the project will investigate whether the improvement is sufficiently large to justify processing the additional data. Conceptually:

$$\text{Additional History} \rightarrow \begin{cases} \text{Additional Recommendation Benefit} \\ \text{Additional Computational Cost} \end{cases}$$

The objective is to identify situations where processing more history provides diminishing returns.

Objective 4 – Develop an adaptive history-management strategy

Based on the previous analysis, design a simple and interpretable mechanism that dynamically determines how much behavioral history should be processed. The system should follow the general loop:

$$\text{Observe} \rightarrow \text{Estimate Benefit/Cost} \rightarrow \text{Adapt History} \rightarrow \text{Recommendation}$$

The adaptive mechanism should avoid processing additional history when its expected benefit is insufficient compared with its cost.

4 Research Methodology

4.1 Dataset

The project will use the KuaiRand short-video recommendation dataset. KuaiRand is suitable for this study because it contains users' historical interactions with short-video content and provides an experimental setting involving random intervention, allowing recommendation performance to be evaluated under a controlled offline setting.

The initial development and experimentation will be conducted on KuaiRand-1K. If the methodology proves feasible and computational resources permit, the final experiments may be extended to KuaiRand-27K.

The primary prediction target is: `long_view`, representing whether the user exhibits a long-view interaction with the recommended video.

4.2 Step 1 – Fixed-history baseline

The first stage will establish a controlled baseline by training/evaluating models with different fixed amounts of behavioral history:

$$K \in \{0, 1, 5, 10, 20, 50, 100, \text{Full}\}$$

For each K, the same general model architecture and evaluation protocol will be maintained so that the main experimental variable is the amount of behavioral history.

The primary recommendation-quality metric will be ROC-AUC. Additional metrics such as log loss may be considered where appropriate. This experiment answers: *How much recommendation performance can be obtained from different amounts of behavioral history?*

4.3 Step 2 – Quality–cost analysis

For every history length K, both recommendation quality and computational cost will be measured. The experiment will therefore produce a relationship of the form:

$$K \rightarrow (\text{Recommendation Quality}, \text{Computational Cost})$$

For example:

History	Recommendation Quality	Data Processed	Latency	Memory
$K = 0$	measured	measured	measured	measured
$K = 1$	measured	measured	measured	measured
$K = 5$	measured	measured	measured	measured
$K = 10$	measured	measured	measured	measured
$K = 20$	measured	measured	measured	measured
$K = 50$	measured	measured	measured	measured
$K = 100$	measured	measured	measured	measured
Full	measured	measured	measured	measured

The purpose is not simply to identify the configuration with the highest recommendation quality, but to characterize the quality–cost trade-off.

4.4 Step 3 – Marginal benefit and cost analysis

The next step will examine the marginal effect of increasing history length. For two history configurations K_1 and K_2:

$$\Delta Q = Q(K_2) - Q(K_1)$$

represents the additional recommendation benefit, while

$$\Delta C = C(K_2) - C(K_1)$$

represents the additional computational cost.

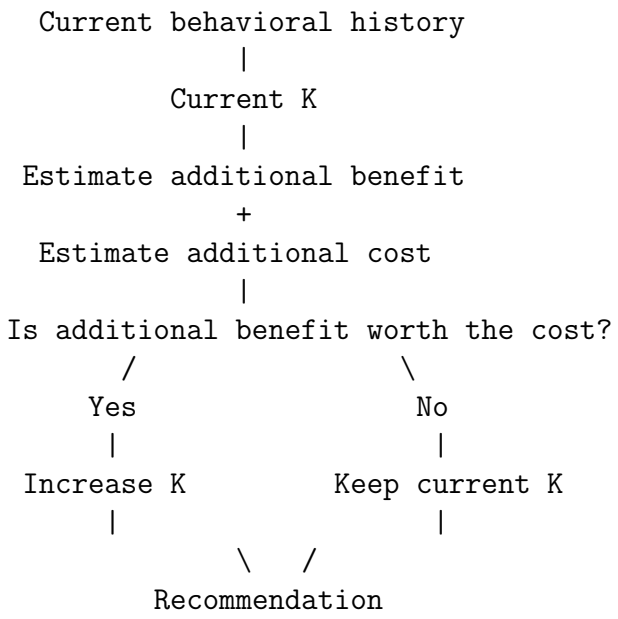
The project will investigate the relationship:

$$\frac{\Delta Q}{\Delta C}$$

as an indication of the additional recommendation benefit obtained per unit of computational cost. This analysis is important because a larger history may improve recommendation quality while providing progressively smaller marginal benefits.

4.5 Step 4 – Adaptive history management

Based on the previous analysis, an adaptive strategy will be designed. Rather than always using a fixed K, the system will determine whether additional history should be processed. A conceptual decision process is:



The initial adaptive strategy will prioritize simplicity and interpretability rather than introducing a complex learning-based controller. For example, the system may progressively increase the history length only when the expected improvement from additional history is sufficiently large relative to its computational cost. More sophisticated adaptive mechanisms may be considered only after the basic cost-benefit relationship has been established.

5 Evaluation

The adaptive strategy will be compared with fixed-history baselines. The main comparison will consider:

- **Recommendation quality:** ROC-AUC, potentially log loss.
- **Computational cost:** inference latency, memory usage, amount of behavioral data processed, computational workload.
- **Resource efficiency:** quality achieved per unit of computational cost, optional energy consumption if reliable measurement is available.
- **Adaptation effectiveness:** reduction in processed history/data, computational-cost reduction, recommendation-quality degradation (if any).

The desired outcome is not necessarily to maximize recommendation quality alone. Instead, the objective is to determine whether the adaptive strategy can achieve a favorable point in the quality–cost trade-off, for example by substantially reducing computational cost while maintaining recommendation quality within an acceptable range of the fixed-history baseline.

6 Relationship to EDAM

The project can be viewed as an EDAM-oriented data-management problem rather than solely a recommendation-modeling problem. The central idea is:

Data-management decision → Amount of data processed → Computational resource usage → System p

In this project, the data-management decision is: *How much behavioral history should be processed for a recommendation request?* A longer history means more behavioral data must be accessed, represented, and processed. However, using less history may reduce recommendation quality. Therefore, behavioral history becomes a data-management resource that can be dynamically controlled according to its expected value and computational cost.

This is also where the BLOOM-FaaS paper provides methodological inspiration. Its early-filtering approach demonstrates the general principle of reducing unnecessary data before expensive computation, while its adaptive coordinator uses runtime information to adjust processing decisions. The paper also highlights that filtering itself introduces overhead and therefore should be applied when the expected benefit justifies the cost.

The proposed project transfers this general principle to a different application domain:

Early data filtering in serverless joins → Adaptive behavioral-history management in recommendation

The project does not aim to reproduce or directly apply Bloom Filters to recommendation. Instead, it adopts the broader ideas of adaptive data reduction, feedback-driven decision making, and cost-benefit-aware processing.

7 Expected Contributions

The project is expected to provide the following contributions:

1. **Empirical characterization of behavioral-history efficiency:** Quantify how much recommendation information can be obtained from different amounts of behavioral history in the KuaiRand environment.
2. **Quality–cost analysis:** Establish the relationship between behavioral-history length, recommendation quality, and computational cost.
3. **Cost-benefit criterion for additional history:** Investigate when additional behavioral history provides enough recommendation benefit to justify its additional processing cost.
4. **Adaptive history-management strategy:** Develop and evaluate an interpretable mechanism that dynamically selects the amount of behavioral history to process.
5. **EDAM-oriented case study:** Demonstrate how a data-management decision in a recommendation workload can be connected to computational-resource efficiency.

8 Scope and Limitations

The project will focus on:

- short-video recommendation;
- behavioral interaction history;
- history-length management;
- recommendation quality;
- computational/resource cost;
- adaptive data-management decisions;
- the KuaiRand experimental environment.

The project will not focus on:

- developing a new recommendation architecture;
- building a production-scale recommendation platform;
- full online A/B testing;
- causal claims about real-world recommendation policies;
- reproducing the BLOOM-FaaS architecture;
- directly applying Bloom Filters to the recommendation model.

The conclusions will therefore be interpreted within the experimental setting and dataset used in the study. Energy consumption will only be included as a primary result if reliable measurement infrastructure is available. Otherwise, computational cost will be evaluated using measurable proxies such as latency, memory usage, workload, and amount of data processed.

9 Expected Hypothesis

The project investigates the following hypothesis:

Increasing behavioral-history length can improve recommendation quality, but the marginal benefit of additional history eventually decreases while computational cost continues to increase. Therefore, an adaptive history-management strategy can potentially reduce computational cost while maintaining an acceptable level of recommendation quality.

This hypothesis will be tested empirically rather than assumed to hold for all users, models, or workloads.

10 Preliminary Experimental Plan

Phase 1 – Dataset and experimental setup: Finalize KuaiRand data pipeline. Verify evaluation data and leakage controls. Define behavioral-history construction. Establish reproducible training/evaluation protocol.

Phase 2 – Fixed-history experiments: Train/evaluate $K \in \{0, 1, 5, 10, 20, 50, 100, \text{Full}\}$. Measure recommendation quality. Measure data-processing and computational cost.

Phase 3 – Quality–cost analysis: Construct quality–cost curves. Analyze marginal benefit of additional history. Identify diminishing-return regions. Investigate conditions under which additional history is worthwhile.

Phase 4 – Adaptive strategy: Design an interpretable adaptive history-selection mechanism. Implement the decision loop. Evaluate different decision thresholds/criteria.

Phase 5 – Evaluation and comparison: Compare adaptive history management against fixed-history baselines. Evaluate quality, latency, memory, workload, and data reduction. Evaluate energy consumption if reliable measurement is available.

Phase 6 – Analysis and documentation: Analyze results. Discuss limitations. Relate findings to EDAM and adaptive data management. Prepare the final report and presentation.

11 Expected Outcome

The final outcome will be a reproducible experimental study demonstrating whether behavioral history can be treated as an adaptively managed data resource in short-video recommendation.

The key result will be a characterization of the relationship:

$$\boxed{\text{Behavioral History} \rightarrow \text{Recommendation Quality} \leftrightarrow \text{Computational Cost}}$$

and an adaptive strategy that attempts to operate near a favorable quality–cost trade-off rather than always processing the maximum available history.

This provides a concrete case study for the broader EDAM objective of making data-management decisions with awareness of their computational and resource implications.